When Small Changes in Words (Prompts), Robot Change Actions

A clear draft paper based on completed OpenVLA experiments

submission draft for CoRL-style packaging

Abstract

Vision-language-action models connect images, language, and robot control in a single model. This makes them powerful, but it also raises a simple safety question: will a robot choose a different action if the instruction wording changes while the scene stays the same? We study this question on OpenVLA-7B. We run two completed evaluations. The first uses a manually curated set of 40 BridgeData-style trajectories with initial and final frames. The second uses an automatically sampled set of 50 scripted BridgeData trajectories. For each image, we compare normal, paraphrased, contradictory, and neutral prompts and measure action drift in the first six action dimensions using L2 distance and cosine similarity. Across both experiments, paraphrases stay closest to the original prompt, while contradictory and neutral prompts usually cause larger drift. Final-state images are generally more fragile than initial-state images. These results show that prompt wording can measurably change OpenVLA actions even when the image is fixed. We conclude that prompt robustness should be treated as a robotics safety problem, and we discuss how future work can connect drift to task success and use disagreement as an inference-time safety trigger. [1][3][4][5][6][7][8][9]

1. Introduction

Vision-language-action (VLA) models aim to turn perception, language understanding, and control into a single robot policy. That promise is exciting because it suggests a future where a person can describe a task in everyday language and a robot can act on it. At the same time, it creates a new safety question: if a user changes the wording slightly, does the robot still behave consistently? Recent papers show that this question is no longer hypothetical. VLA systems can show weak language grounding, contradiction failures, and safety problems that are hidden by visually easy benchmarks. [1][4][5][6][7][8][9]

This paper studies that question on OpenVLA-7B. Our goal is narrow and practical. We do not claim to solve language robustness for all VLAs. Instead, we ask whether small wording changes change OpenVLA actions, and whether the effect looks different at the beginning and end of a task. We test this on two completed experiment branches: Manual40 and Auto50. The first is a hand-curated evaluation with 40 trajectory pairs. The second is an automatically sampled validation branch with 50 trajectory pairs. In both branches, we compare normal, paraphrased, contradictory, and neutral prompts.

Our main findings are simple. First, paraphrases produce the smallest action drift on average. Second, contradictory and neutral prompts produce larger drift. Third, final-state images are more fragile than initial-state images in both experiment branches. These findings are not yet a task-success benchmark, but they provide a reproducible, model-level warning signal that can motivate future safety mechanisms.

2. Related work

Recent work has made VLA robustness a fast-moving area. The 2026 VLA survey organizes the field across concepts, training, applications, and challenges. LeRobot provides open infrastructure that lowers the barrier to reproducible robot learning. OpenVLA made open-source VLA inference practical for many researchers. [1][2][3]

The closest technical neighbors to our study are recent papers on language grounding and safety. LangGap shows that strong benchmark scores can coexist with weak language understanding. IGAR proposes a train-free way to reduce contradiction failures at inference time. BayesianVLA argues that data collection can create information collapse, where the instruction becomes predictable from vision and the model starts to ignore language. SafeVLA studies safety alignment through constrained learning. SAFE studies general failure detection for VLA policies. FailSafe studies reasoning and recovery from failures. [4][5][6][7][8][9]

Our work differs in scope. We do not introduce a new benchmark, a training method, or a recovery model. Instead, we provide a compact, reproducible action-drift study on OpenVLA-7B that compares paraphrase, contradiction, and neutral prompt families across initial and final task states, and we validate the pattern on both a manual and an automatic dataset branch.

3. Method

We run OpenVLA-7B locally in 4-bit mode under WSL on an RTX 2060 GPU. During setup, the Hugging Face inference path showed an attention-mask mismatch between image tokens and prompt tokens. We patched predict_action by prepending a 256-token image mask before generation, which stabilized local inference.

For each evaluated image, we construct four prompts: normal, paraphrased, contradictory, and neutral. We then compute the predicted action and compare the first six action dimensions against the normal prompt. We use L2 distance and cosine similarity. Lower L2 means smaller drift. Higher cosine means stronger directional agreement.

Manual40 contains 40 initial images and 40 final images from a hand-curated trajectory set. Auto50 contains 50 initial images and 50 final images from an automatically sampled scripted BridgeData subset. Manual40 emphasizes control and interpretability. Auto50 emphasizes broader validation.

4. Results

Table 1 shows the main summary metrics. In Manual40 initial frames, average L2 drift is 0.028126 for paraphrases, 0.042365 for contradictory prompts, and 0.045330 for neutral prompts. In Manual40 final frames, these values increase to 0.043141, 0.059920, and 0.070772. In Auto50 initial frames, they are 0.056164, 0.069716, and 0.073405. In Auto50 final frames, they are 0.080668, 0.093857, and 0.082103. The consistent pattern is that paraphrases stay closest to the original prompt, while contradictory and neutral prompts move farther away. Final-state images are also more fragile than initial-state images.

The count analysis supports this pattern. In Manual40, paraphrase is the closest prompt family in 25/40 initial cases and 22/40 final cases. In Auto50, paraphrase is closest in 27/50 initial cases and 26/50 final cases. Some near-invariant cases also exist, but they are rare and do not erase the overall trend.

These results matter because they show that action drift can be observed without changing the image at all. Only the wording changes. That means language robustness is not a side issue. It is part of safe robot control.

5. Discussion, limitations, and future work

This study has several limits. It focuses on one main VLA model, OpenVLA-7B. It measures action drift rather than end-task success. It also works from still images rather than full closed-loop trajectories. These are meaningful limits and they lower the strength of any claim about real robot failure.

At the same time, the work is useful because it creates a reproducible warning sign. A natural next step is to test whether high-drift cases fail more often in simulation or in replay-style evaluation. Another next step is to turn drift into a safety trigger: if alternate prompts disagree too much with the normal prompt, the system can re-prompt, abstain, or retreat to a safe state. These extensions would move the work from diagnosis toward mitigation, which is likely a stronger path for a conference submission.

Why has the problem not already been solved? One reason is dataset shortcutting. In many robot datasets, the image alone already gives away much of the task, so the model can ignore language and still look strong on standard benchmarks. Another reason is evaluation cost. Real robot and simulator studies take time, engineering, and careful safety design. A third reason is systems complexity: modern VLAs already sit at the edge of what many student-scale setups can run. [4][5][6][7][8][9]

6. Conclusion

We presented a compact action-drift study for OpenVLA-7B across two completed evaluation branches, Manual40 and Auto50. In both branches, paraphrased prompts stayed closest to the original prompt, while contradictory and neutral prompts caused larger drift. Final-state images were generally more fragile than initial-state images. These results suggest that prompt wording can change robot actions even when the visual scene stays fixed. We believe prompt robustness should be treated as a robotics safety issue, and that future work should connect action drift to task success and add inference-time mitigation.

Main table

Experiment	Group	n	L2 paraphrase	L2 contradiction	L2 neutral	Cos paraphrase	Cos contradiction	Cos neutral
manual40	initial	40	0.028126	0.042365	0.045330	0.678776	0.586046	0.425675
manual40	final	40	0.043141	0.059920	0.070772	0.480281	0.219040	-0.001856
auto50	initial	50	0.056164	0.069716	0.073405	0.483856	0.269209	0.170514
auto50	final	50	0.080668	0.093857	0.082103	0.386891	0.157202	0.220440

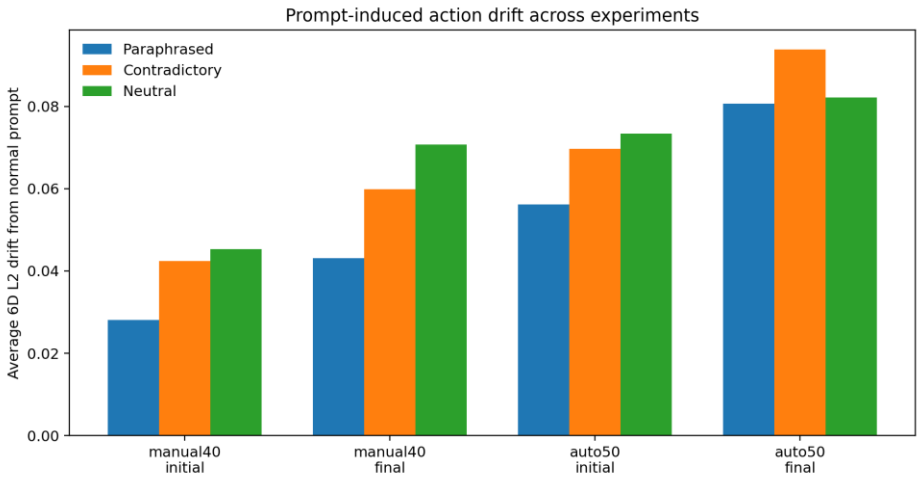

Figure 1. Summary view of average drift.

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